

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Vinod Gupta School of Management
2. **Subject Name:** Financial Analytics **L-T-P:** 0-0-3 **Credits:** 2
3. **Pre-requisites:** Corporate Finance and Financial Accounting
4. **Syllabus and reference books:**

Syllabus:

The objective of this course is to teach valuation of various financial instruments like equity, bonds, options, and analyse the econometric methods used for financial data. Largely the course deals with the time series and panel data methods.

Reference Books:

- 1) Chandan Sengupta, Financial Modeling. Wiley Finance

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to the theories of asset pricing, Capital asset pricing model (CAPM), Arbitrage pricing theory (APT), Consumption CAPM, Factor models, Efficient frontier, Optimal portfolio selection	12
2.	Valuation of firms, Issues related to the valuation of equity versus that of firms, Difficult-to-value firms, Valuation for IPOs	6
3.	Fixed income modelling, Valuation, duration, convexity, fixed income portfolios, Option pricing using binomial method and Monte Carlo simulations	6
4.	Maximum likelihood estimation, Time-series modelling, AR, MA, ARMA, ARIMA, ARCH, GARCH, VAR, VECM	6
5.	Bankruptcy prediction using classification models, Logistic regression, ANN, SVM, RF, Text analysis	6
Total number of hours		36